



THERMAL COMFORT IN OUTDOOR URBAN SPACES: UNDERSTANDING THE HUMAN PARAMETER

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Abstract—The research being undertaken seeks to achieve a better understanding of the richness of microclimatic characteristics in outdoor urban spaces, and the comfort implications for the people using them. The underlying hypothesis is that these conditions influence people's behaviour and usage of outdoor spaces. The initial results demonstrate that a purely physiological approach is inadequate in characterising comfort conditions outdoors, and an understanding of the dynamic human parameter is necessary in designing spaces for public use. The thermal environment is indeed of prime importance influencing people's use of these spaces, but psychological adaptation (available choice, environmental stimulation, thermal history, memory effect, expectations) is also of great importance in such spaces that present few constraints. © 2001 Elsevier Science Ltd. All rights reserved.

1. INTRODUCTION

The initial findings of the research being presented here seek to achieve a better understanding of the richness of microclimatic characteristics in outdoor urban spaces and the comfort implications for the people using them. The underlying hypothesis is that these conditions influence people's behaviour and usage of outdoor spaces. Responses to the microclimate may be unconscious, but they often result in a different use of urban space in different climatic conditions.

Improved microclimatic conditions have in turn major implications for the development of cities. By controlling sources of discomfort, sedentary activities, as well as the use of public transport, cycling and walking, are promoted. Successful areas attract large numbers of people, which in turn attract businesses, workers, residents, and the area becomes economically profitable. The energy use of the surrounding buildings can also be affected. Finally successful outdoor spaces can benefit the image of the city.

This has been recognised in San Francisco, where by legislation, new buildings have to be sited so that winds at ground level are limited to 5 m/s in areas where people sit for 90% of daylight hours. Also new construction cannot cast a shadow on a park between one hour after sunrise and one hour before sunset, throughout the year (Bosselmann *et al.*, 1988).

1.1. Urban design and outdoor spaces

There is adequate understanding of the influence of climate on urban settlements. For example, the way self-shading streets protect the buildings and the surrounding spaces from the hot sun, in hot arid climates; or dispersed buildings to allow for easy flow of wind through the spaces in hot humid climates etc.; as well as problems created by importing architectural vocabulary without adapting it for the local climate is currently acknowledged.

In cases where the urban fabric is existent, however, alterations are possible down to the scale of the urban block to improve the microclimate. Examples demonstrate that urban microclimatic control is possible. Shading whole streets is feasible whether in the form of trees, with the advantage of cooling via evapotranspiration through the leaves, or man-made canopies with the available materials, solutions found worldwide and not unique to specific cultures.

1.2. Thermal comfort

Conventional comfort theory relies on a steady-state model where the production of heat is equal to the heat losses to the environment, aiming to keep a constant core body temperature of 37°C, so that the environmental conditions which provide thermal satisfaction, dependent only upon the activity of the subjects and their clothing level, fall within a narrow band.

Subsequently work showed that people take action to improve their comfort conditions by modifying their clothing and metabolic rate, or by interacting with the building, 'adaptive' actions

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(Nicol, 1990). Separating thermal sensation from thermal satisfaction, it was further demonstrated that 'adaptive opportunity' (the degree to which people can adapt to their environment) is important for their satisfaction with the space. Conversely when adaptive opportunity is limited, departure from neutrality causes stress and dissatisfaction (Baker and Standeven, 1996). This suggests that intrinsic factors like past experience (Hawkes and Willey, 1977), naturalness (Griffiths *et al.*, 1987), experience and expectations (de-Dear, 1995), time of exposure (Baker, 1993), and the need for environmental stimulation (Givoni, 1976; Baker and Standeven, 1996) are also important for thermal satisfaction.

1.2.1. Comfort conditions outdoors. Due to the great complexity of the outdoor environment, in terms of variability, temporally and spatially, as well as the great range of activities people are engaged in, there have been very few attempts to understand comfort conditions outside.

The first people to tackle such issues were Gold in the 1930s and Siple and Passel in the Antarctic in the 1940s, the creators of the wind-chill index. In the absence of any other similar studies, forty years later Penwarden (1973) attempted a more systematic approach for thermal conditions outdoors, by adding to the steady state model, a term for solar radiation.

In the 1980s a team of researchers at Berkeley (Bosselmann *et al.*, 1984) worked on thermal comfort outdoors, particularly on implications of design solutions for the microclimate of San Francisco, which led to the legislation for solar access and wind protection. Again a mathematical model of the thermoregulatory system was employed for calculating the comfort conditions.

2. RESEARCH FRAMEWORK

In all the attempts to understand the thermal conditions outdoors, a purely physiological model has been used, similar to the model used for the indoor environment, adapted for the solar radiation parameter. However, there is a lack of understanding of the human parameter in these spaces, and their subjective responses.

If 'adaptive opportunity' is significant enough in order to call for a change of the comfort standards used by the building services' profession, what would be the situation outdoors, where the environment is unconstrained and the only modification is action taken by people?

2.1. Case studies

Investigating the comfort conditions of open urban spaces, a differentiation has to be drawn between routes and resting places. In this work we are concerned with resting areas, as people choose to sit somewhere, whereas they do not necessarily choose a particular route, in order to avoid discomfort. Nevertheless this discomfort will not cause them serious stress, since the time of exposure to the specific environmental conditions is short. On the other hand, with resting places the situation is different, as poor comfort conditions may distress people and lead them to avoid using these areas.

Four case study sites have been identified in the city-centre of Cambridge, which have been designed specifically for external public use. They can be identified as urban squares, streets or parks, with strong commercial activities and very popular in terms of public use. The sites varied in terms of typology, geometry, orientation and intended use, but are all set in the urban context and are used as resting places apart from thoroughfares. The field studies took place during spring, summer and winter 1997. A total of 1431 people were interviewed, and this figure would have been higher, if only there had been more people sitting outdoors during winter. One week was spent at each site, each season (2 h at lunchtime, 3 weekdays and the weekend, to get the weekly pattern of use, since weekdays tend to be quite different from the weekends).

A portable mini-met station was placed next to the interviewee who was always sedentary, to record the objective environmental parameters the person was experiencing (such as air temperature, solar radiation, wind and humidity). These were then compared with subjective behaviour and responses (from structured interviews and observations) to evaluate the thermal comfort conditions people experience. Individuals' characteristics — as well as behavioural ones — such as age, sex, race, clothing, posture and activity, were taken into account. Some of the initial results are presented below.

2.2. Use of outdoor space

The first issue investigated was whether the thermal and, by implication, comfort conditions, affect people's use of outdoor spaces. The simplest way to examine that was by counting the number of people using the spaces at various intervals, during the interview period, a mean

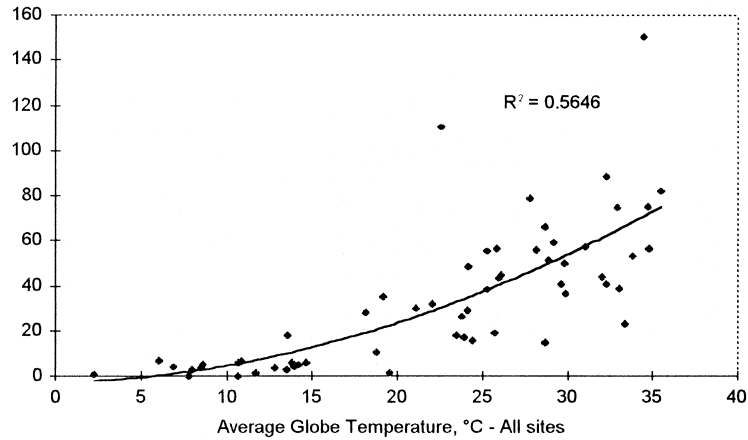


Fig. 1. Variation of the number of people outdoors, in relation to globe temperature for all the sites.

value eventually calculated for each day. It is clear that warm conditions and presence of sunlight are important factors in the use of the space. Fig. 1 presents the average number of people

sitting in the space increasing as global temperature increases.

Disaggregating the results for the different sites (Figs. 2–5), the situation is similar, apart from

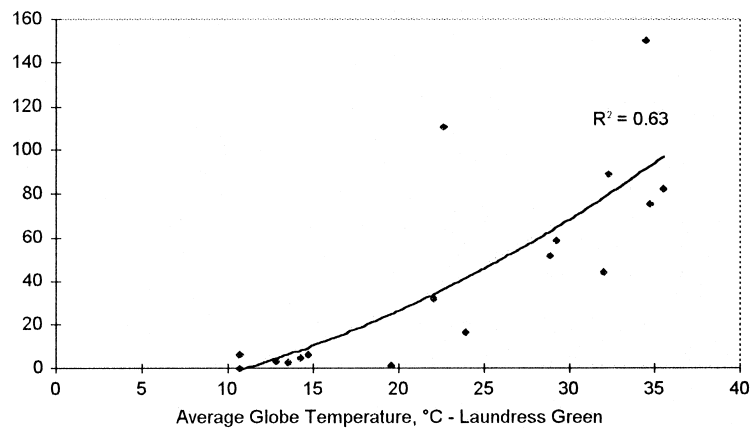


Fig. 2. Variation of the number of people outdoors, in relation to globe temperature for Laundress Green.

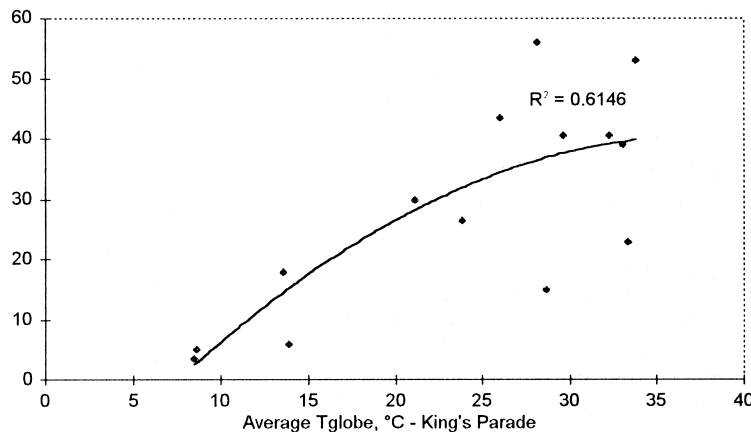


Fig. 3. Variation of the number of people outdoors, in relation to globe temperature for King's Parade.

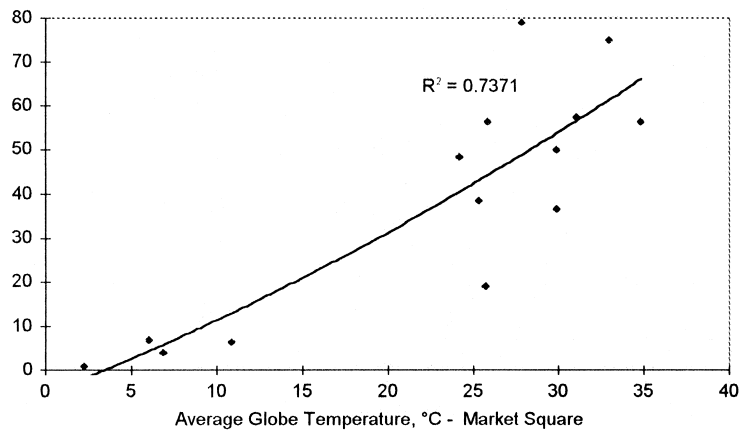


Fig. 4. Variation of the number of people outdoors, in relation to globe temperature for Market Square.

one site, King's Parade (Fig. 3), where the curve, although of increasing slope, tends to stabilise after about 25°C. All the other three sites have a variety of spaces with sun and shade, where people have a wide choice of spaces to sit, under different conditions. King's Parade is a long narrow street running N–S, with a 0.5 m wall along the street, where people sit. There are however, no means of shading available, and it is in sun for the most part of the day. This is much appreciated during the cooler season, but in the summer, it presents problems. Less people are therefore using the space, and because there is a general lack of sitting places available in the city-centre, people sit, but for shorter periods than they would do otherwise, e.g. for some rest, to finish off their lunch etc.

2.3. Thermal sensation

The interviewees were reporting their thermal sensation and value judgment, on a 5-point scale,

varying from too cold to too hot. Examining the Actual Sensation Vote in terms of frequencies for the whole year, as well as the different seasons separately (Fig. 6), it is interesting to notice that the highest number of frequencies is for +1, with over 50% of the votes. This is similar for spring and summer, but for winter it changes to –1. The slight bias towards warm temperatures partially explains that.

To compare the different seasons on an equal base, not taking into account that the number of people for each season was different, percentages are presented instead (Fig. 7). The extreme ± 2 have very few occurrences for the whole year with 13%, whereas the seasonal results vary. In the summer +2 is higher with 20%, whereas –2 in winter is 12%. For winter there are no +2 votes which is understandable, since air temperatures were low, whereas for the summer there is a 2% for –2. In spring, where thermal conditions are milder the extremes are low, 6% for –2 and

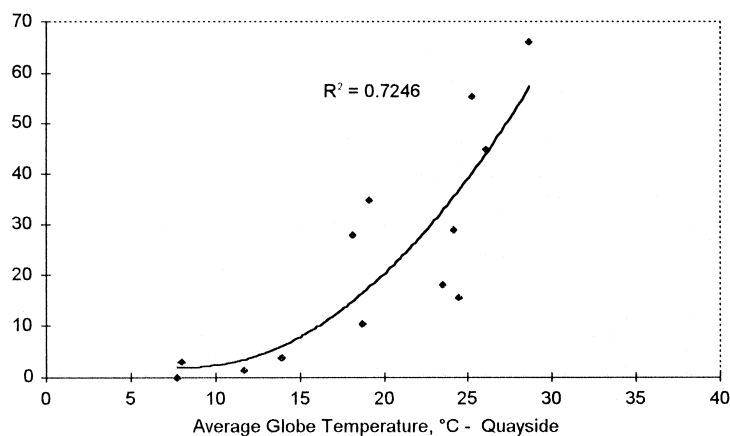


Fig. 5. Variation of the number of people outdoors, in relation to globe temperature for Quayside.

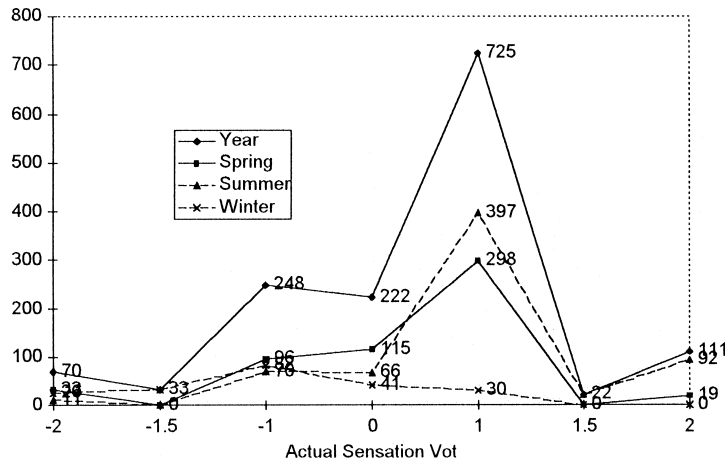


Fig. 6. Frequency distribution for the Actual Sensation Vote for the different seasons as well as combined for the whole year.

3% for -2 . Another interesting point that surfaces, is that the number of people voting for 0, i.e. neutrality, is not very high, varying from 10 to 20%, ± 1 having higher frequencies. This seems to suggest that the majority of people are comfortable feeling warm or cool, as opposed to neutral, enjoying the stimulation thermal conditions outdoors provide.

Furthermore, the fact that the maximum number of people outside is found in the summer, when the maximum number of thermal votes is falling in the $+1$ category (Fig. 6), suggests that most people enjoy feeling warm. It could therefore be speculated that in warmer climates the situation would be reversed and more people would be found outdoors when temperatures would be lower than average and the majority of votes would fall on the cool side, e.g. in the evenings when the sun has set.

Comparison of objective and subjective data

was of great importance. Therefore, the Actual Sensation Vote was then compared with the corresponding theoretical Predicted Mean Vote (ISO 7730, 1994), taking into account the mean objective environmental parameters recorded for the duration of the interview, clothing levels and metabolic rate, for each interviewee. The percentage frequency distribution combined for all the three seasons is shown in Fig. 8.

It is apparent that there is a great discrepancy between the two curves. More than 50% of the people have voted for the warm part of the scale, around 20% for the cool, whereas for the extreme ± 2 , the value is around 10%. The corresponding theoretical PMV curve is very different and much flatter, the hatched area representing the difference between the area underneath the PMV lying outside the Actual Sensation Vote (ASV) curve. Only 35% of the interviewees are within the acceptable theoretical comfort conditions, the vast

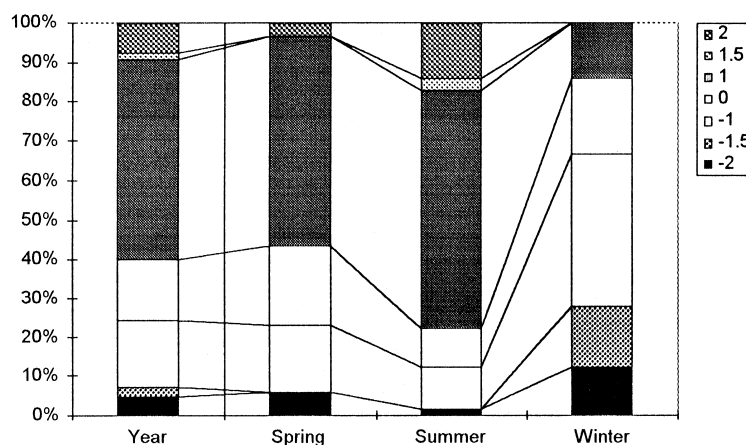


Fig. 7. Percentage frequency for the Actual Sensation Vote.

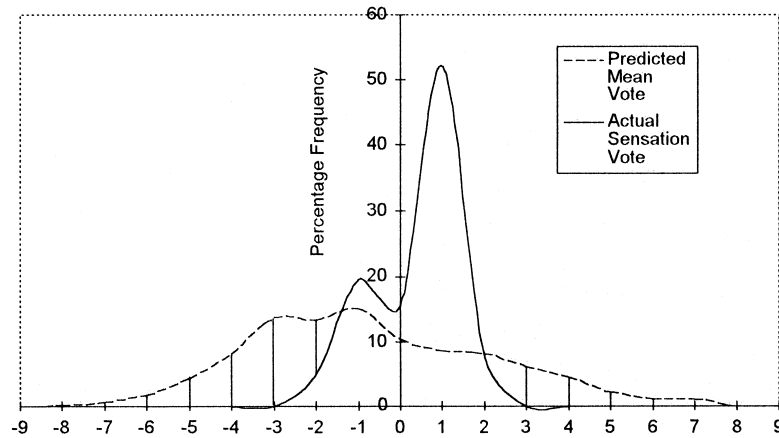


Fig. 8. Percentage frequency distribution for Predicted Mean Vote and Actual Sensation Vote.

majority lying in conditions too hot or too cold, since the PMV calculated for the conditions people have experienced lie from -9 to $+7$!

Presenting this as a measure of dissatisfaction of people for the various conditions (Fig. 9), we notice that the Predicted Percentage of Dissatisfied (ISO 7730, 1994) (getting a mean value from the individual calculated PPDs), varies from 56% in spring to 91% in winter, and the yearly average is 66%. That implies 944 of the 1431 people sitting outside should be dissatisfied with their thermal environment. In fact the Actual Percentage of Dissatisfied (APD) is always around 10%, a figure that is regarded as acceptable, found even in controlled indoor environments.

These results suggest a number of things, the most apparent, that a purely physiological approach is inadequate to characterise thermal comfort conditions outdoors. Although physical adaptation takes place, i.e. with the seasonal variation of clothing (Fig. 10), and the increased

consumption of cool drinks noticed with increasing temperatures and presence of sunlight (correlation coefficient $r = 0.42$ and 0.61 respectively, for $P < 0.01$), the results of the interviews imply that psychological adaptation seems to be more important, as reinforced by the following figures.

2.4. Thermal neutrality

This term was first introduced by Humphreys (1975), when he showed that the variation of the neutral temperature is associated with the variation of the mean temperature. To examine the relationship prevailing in outdoor spaces, an average neutral temperature (T_n) was calculated for each interview day (by getting an average of the calculated T_n for each interviewee) and plotted as a function of the meteorological mean outside air temperature for the previous month (Fig. 11). Thermal neutrality for outdoor conditions was found to vary from 7.5°C in winter to 27°C in the summer, a range of 20°C , whereas the range

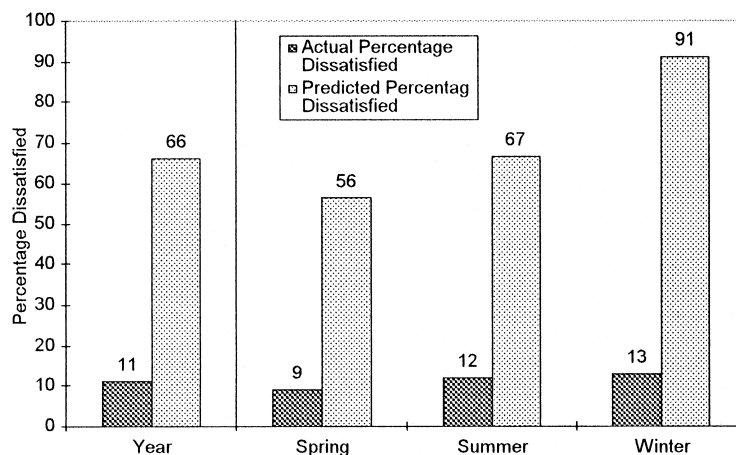


Fig. 9. Comparison between Actual Percentage Dissatisfied and Predicted Percentage Dissatisfied.

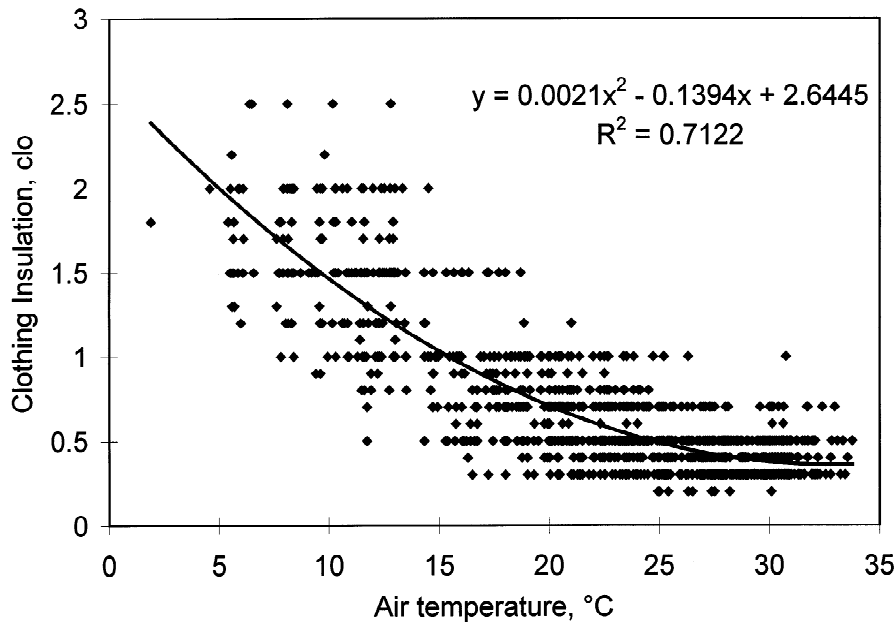


Fig. 10. Variation of clothing insulation with air temperature.

Humphreys had in the indoor environment was 13°C. Seasonal compensation of clothing (Fig. 10), and small changes to the net metabolic heating with the consumption of cool or hot drinks partly justify these extra 7°C. The strongest reason though is psychological adaptation.

Humphreys originally suggested that the fact that T_n lies close to the mean T_{air} is an indication of the influence of the subject's recent experience. Fig. 11 shows that Humphreys' suggestions seem to apply outdoors too, with *memory* of recent conditions being indeed a very critical parameter on the satisfaction with the thermal environment. *Expectations* have an important role in two ways. Firstly they may influence a person's clothing

choice which will effect their actual thermal sensation. Secondly, due to 'psychological preparation' it may influence their interpretation of that sensation as dissatisfaction. T_n is always higher than Mean T_{air} (which does not exceed 21°C), showing that people would prefer it were warmer, although they may have partially compensated by clothing etc. This is in agreement with McIntyre's findings (McIntyre, 1980), where he showed that people voting for the neutral temperatures in warm climates prefer to be cool and people in cold climates prefer to be warm.

To identify whether short-term memory is a better indicator of T_n than long-term memory (Humphreys and Nicol, 1995), T_n was plotted

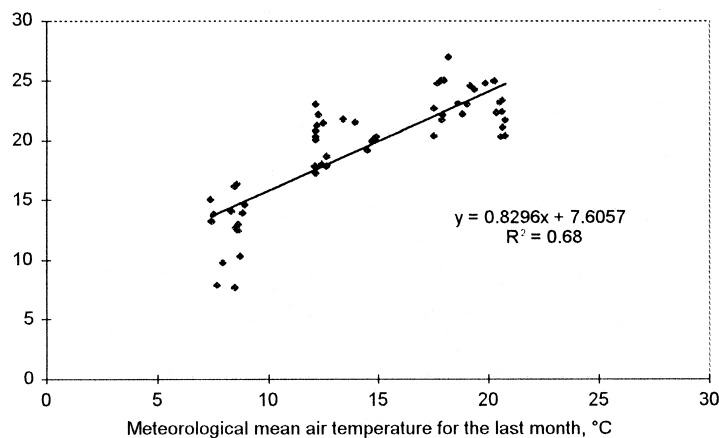


Fig. 11. Neutral temperature as a function of air temperature for the past month.

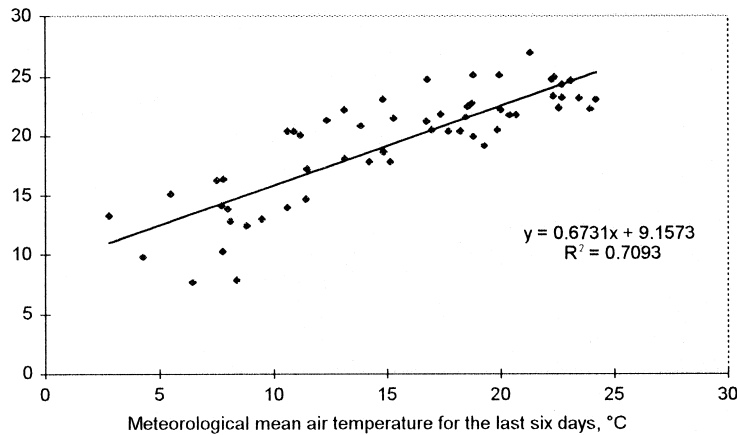


Fig. 12. Neutral temperature as a function of air temperature for the past week.

against the meteorological mean outside T_{air} for the last six days (Fig. 12). This illustrates a slightly improved linear correlation between the two variables ($r = 0.84$), suggesting that short-term memory is indeed an even better indicator of T_n , than long-term memory.

When T_n is plotted against current meteorological T_{air} (for the duration of the interview), the correlation improves even more (Fig. 13). However, this does not indicate that there is no memory effect, since there is a strong correlation between hot days and warm months. Furthermore, current temperature influences the current dress state and other adaptive mechanisms available, e.g. position, posture and consumption of cool or hot drinks to alter the net metabolic heating, as opposed to the memory effect we get in the previous Figures. It is interesting to notice that T_n is again higher than Mean T_{air} for low temperatures, become roughly equal for the range 17–

24°C, and above that becoming lower than T_{air} , where people would prefer it were cooler.

3. CONCLUSIONS

Outdoor spaces present few constraints. People sit there by their own free choice. They are there because they want to feel the warm sun rays and fresh air, and look at other people, while they are having their lunch for example. *Environmental stimulation* is therefore important. The degree to which they want to ‘charge up’ their body with heat and fresh air is important, especially when considered in combination with a person’s *thermal history*, i.e. whether before they were working in an air-conditioned office. The important role of *memory* and *expectations* has already been mentioned, both in terms of physical and psychological adaptation.

It is quite tempting to assume from the above

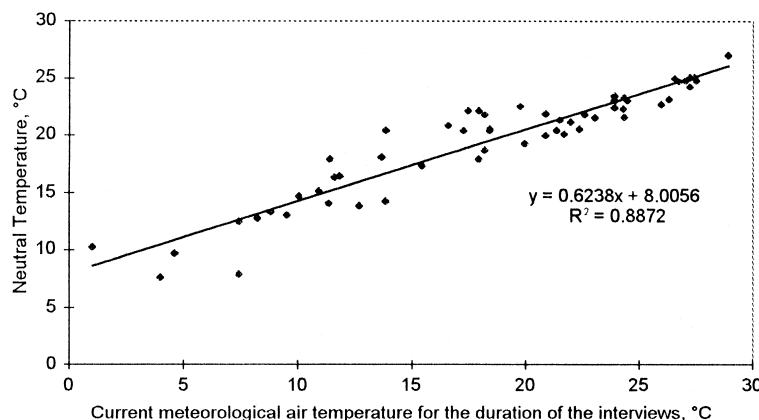


Fig. 13. Neutral temperature as a function of current meteorological mean air temperature for the duration of the interviews.

that the physical environment per se is not important in outdoor comfort, and that other parameters are more important in creating comfortable spaces. This is not necessarily so. As Figs. 1–5 show the first level of importance is to get the people to come out of the buildings they are working in, or to stop them while they are passing through the space, going elsewhere, and this is a direct implication of the thermal environment.

What Figs. 6–9 and 11–13 demonstrate is the second level of importance. Once people have decided to come out and sit outside, they have accepted various facts about the thermal conditions and are prepared to accommodate them. That is where the importance of psychological adaptation lies.

Clearly a purely physiological approach is inadequate in characterising thermal comfort conditions in outdoor spaces. The design of open spaces is very important for the urban environment and an understanding of the wide range of effects influencing thermal comfort in these spaces will assist in designing spaces encouraging public use, at different times of the year.

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